→ Ex:-6:- Let's go with small example:- what is highest power of 3 when it devides 10!?

Ans:- $\frac{10!}{3^x} \Rightarrow x = ? \quad \frac{1\times2\times3\times4\times5\times6\times7\times8\times9\times10}{3^?}$

Means Numerator is to be written in terms of $NO\times3^n$.

$$\Rightarrow \frac{1\times2\times\textcircled{3}\times4\times5\times(2\times\textcircled{3})\times7\times8\times(\textcircled{3}\times\textcircled{3})\times10}{3^?}$$

$$= \frac{1\times2\times4\times5\times2\times7\times8\times\times10\times3^4}{3^?} \quad \text{here } 3^? = 3^4 \Rightarrow \boxed{?=4}$$

(or) Division method:- like LCM division:-

$$\frac{1\times2\times\underset{\downarrow}{\textcircled{3}}\times4\times5\times(2\times\underset{\downarrow}{\textcircled{3}})\times7\times8\times(\underset{\downarrow}{\textcircled{3}}\times\underset{\downarrow}{\textcircled{3}})\times10}{\textcircled{1} \quad 3^? \quad \textcircled{1} \quad \textcircled{1}\ \textcircled{2}}$$

3 | 10
3 | 3 → 3 times 3's in first division of 3 multiples
1 → 1 time 3 in 2nd division which 2nd 3 in '9'.

totally $\boxed{x = ? = 4}$.

↳ When highest power of composite number devides a particular !:—

→ When composite number highest power is asked. in certain ! [like $\frac{600!}{6^x}$]. then we have to convert the composite number into product of factors in such a way that one of them is prime (or) both primes. then do the same division method with bigger prime. Number.

Ex:7:- What is value of x in $\frac{600!}{6^x}$?

Ans:- $6 = 2 \times 3$, in these 2 & 3, 3 is highest prime, so. do the division method with 3 & the resultant is only answer.

$$\begin{array}{r|l} 3 & 600 \\ 3 & 200 \\ 3 & 66 \\ 3 & 22 \\ 3 & 7 \\ & 2 \end{array}$$

$x = 200 + 66 + 22 + 7 + 2 = \boxed{297}$

Ex:- $\frac{10!}{6^x} = \frac{1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \times 9 \times 10}{6^x}$

$= \frac{1 \times (2 \times 3) \times (2 \times 2) \times 5 \times (6) \times 7 \times (2 \times 2 \times 2) \times (3 \times 3) \times 10}{6^x}$

$= \frac{1 \times (2 \times 3) \times (2 \times 2) \times 5 \times (2 \times 3) \times 7 \times 2 \times (2 \times 3)(2 \times 3) \times 2 \times 5}{6^x}$

$= \frac{1 \times 2^4 \times 5^2 \times 7^1 \times (2 \times 3)^4}{6^x} \Rightarrow \boxed{x = 4}$

Note:- Why we have considered highest value (Prime) in the prime factors because that many combinations only we will get [even thought smaller numbers are more, we can't form the combinations].

Ex:-8:- What is value of x in $\frac{200!}{10^x}$ (or)

What is highest power of 10 in 200! (or)

How many zeroes in 200! [$200! = \ldots\ldots \times 10^x$]

Ans:- 10 is composite $\Rightarrow \therefore 10 = 2 \times 5$

$$\begin{array}{r|l} 5 & 200 \\ 5 & 40 \\ 5 & 8 \\ & 1. \end{array}$$

$x = 40 + 8 + 1 = \boxed{49}$

Ex:-9:- What is value of x in $\frac{100!}{8^x}$?

Ans:- 8 is a composite number $\Rightarrow 8 = 2\times2\times2 = 2^3$

So, here we have to do with '2' & devide the resultant quotient with '3'.

2 | 100	$Q = 50+25+12+6+3+1$
2 | 50	$= 97$
2 | 25	
2 | 12	2^{97} but we want 8^x
2 | 6	$\Rightarrow (2^3)^{\frac{97}{3}} = (8)^{32}\cdot 2^1$ $\Rightarrow \boxed{x = 32}$
2 | 3.	
1	

→ Remainder theorem:-(or) Finding Remainders:-

(i) When $\frac{f(x+a)}{f(x)}$ is equation then Rem → $f(+a)$

(ii) When $\frac{f(x-a)}{f(x)}$ is equation then Rem → $f(-a)$.

In the above 2 cases we have to write numerator in terms of denominator.

Ex:- 10:- Remainder of $\frac{8^{43}}{7}$?

Ans:- $\frac{(7+1)^{43}}{7} \Rightarrow$ Rem $= (1)^{43} = \boxed{1}$

Ex:-11:- $\frac{9^{57}}{7}$ Rem → ? $\Rightarrow \frac{(7+2)^{57}}{7}$ → Rem

2^{57} is Rem > 7 then continue till to get $N^r < 7$

$\frac{2^{57}}{7} = \frac{(2^3)^{19}}{7} = \frac{8^{19}}{7} = \frac{(7+1)^{19}}{7} \Rightarrow$

Rem $= (+1)^{19} = \boxed{1}$